

HDOC Day-31 — Matrix Operations Using VLA

Algorithm

1. Start.
 2. Display the menu:
 - Matrix addition — prove commutative.
 - Matrix multiplication — prove not commutative.
 - Exit.
 3. Read the user's choice.
 4. Use `switch` to perform the selected operation.
 5. For **matrix addition**:
 - Input rows and columns.
 - Create two Variable Length Arrays (VLAs), A and B.
 - Input both matrices.
 - Calculate $A + B$.
 - Calculate $B + A$.
 - Compare both resultant matrices.
 - If equal, display that matrix addition is commutative.
 6. For **matrix multiplication**:
 - Input the order of matrices A and B.
 - To calculate both AB and BA , check:
 - columns of A = rows of B
 - columns of B = rows of A
 - Input both matrices.
 - Calculate AB .
 - Calculate BA .
 - Compare the products if they have the same dimensions.
 - Show that generally $AB \neq BA$.
 7. Repeat the menu until the user selects Exit.
 8. Stop.
-

C Program

```
#include <stdio.h>

// Function to input a matrix
void inputMatrix(int r, int c, int a[r][c])
{
    for (int i = 0; i < r; i++)
    {
        for (int j = 0; j < c; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }
}
```

```

    }
}

// Function to display a matrix
void displayMatrix(int r, int c, int a[r][c])
{
    for (int i = 0; i < r; i++)
    {
        for (int j = 0; j < c; j++)
        {
            printf("%d\t", a[i][j]);
        }
        printf("\n");
    }
}

// Function to prove addition is commutative
void matrixAddition()
{
    int r, c;

    printf("Enter number of rows and columns: ");
    scanf("%d %d", &r, &c);

    int A[r][c], B[r][c];
    int sumAB[r][c], sumBA[r][c];

    printf("Enter elements of Matrix A:\n");
    inputMatrix(r, c, A);

    printf("Enter elements of Matrix B:\n");
    inputMatrix(r, c, B);

    // A + B
    for (int i = 0; i < r; i++)
    {
        for (int j = 0; j < c; j++)
        {
            sumAB[i][j] = A[i][j] + B[i][j];
        }
    }

    // B + A
    for (int i = 0; i < r; i++)
    {
        for (int j = 0; j < c; j++)
        {
            sumBA[i][j] = B[i][j] + A[i][j];
        }
    }

    printf("\nA + B:\n");
    displayMatrix(r, c, sumAB);

    printf("\nB + A:\n");
    displayMatrix(r, c, sumBA);
}

```

```

int equal = 1;

for (int i = 0; i < r; i++)
{
    for (int j = 0; j < c; j++)
    {
        if (sumAB[i][j] != sumBA[i][j])
        {
            equal = 0;
        }
    }
}

if (equal)
    printf("\nA + B = B + A\n"
           "Hence, matrix addition is COMMUTATIVE.\n");
else
    printf("\nMatrices are not equal.\n");
}

// Function to prove multiplication is not commutative
void matrixMultiplication()
{
    int r1, c1, r2, c2;

    printf("Enter rows and columns of Matrix A: ");
    scanf("%d %d", &r1, &c1);

    printf("Enter rows and columns of Matrix B: ");
    scanf("%d %d", &r2, &c2);

    // Both AB and BA must be possible
    if (c1 != r2 || c2 != r1)
    {
        printf("\nBoth AB and BA cannot be calculated "
               "with these dimensions.\n");
        return;
    }

    int A[r1][c1], B[r2][c2];
    int AB[r1][c2], BA[r2][c1];

    printf("Enter elements of Matrix A:\n");
    inputMatrix(r1, c1, A);

    printf("Enter elements of Matrix B:\n");
    inputMatrix(r2, c2, B);

    // Calculate AB
    for (int i = 0; i < r1; i++)
    {
        for (int j = 0; j < c2; j++)
        {
            AB[i][j] = 0;

            for (int k = 0; k < c1; k++)
            {

```

```

        AB[i][j] += A[i][k] * B[k][j];
    }
}

// Calculate BA
for (int i = 0; i < r2; i++)
{
    for (int j = 0; j < c1; j++)
    {
        BA[i][j] = 0;

        for (int k = 0; k < c2; k++)
        {
            BA[i][j] += B[i][k] * A[k][j];
        }
    }
}

printf("\nAB:\n");
displayMatrix(r1, c2, AB);

printf("\nBA:\n");
displayMatrix(r2, c1, BA);

// Compare AB and BA
int equal = 1;

if (r1 != r2 || c2 != c1)
{
    equal = 0;
}
else
{
    for (int i = 0; i < r1; i++)
    {
        for (int j = 0; j < c2; j++)
        {
            if (AB[i][j] != BA[i][j])
            {
                equal = 0;
            }
        }
    }
}

if (!equal)
    printf("\nAB != BA\n"
           "Hence, matrix multiplication is NOT COMMUTATIVE.\n");
else
    printf("\nAB = BA for these particular matrices.\n"
           "However, matrix multiplication is not "
           "commutative in general.\n");
}

int main()
{

```

```

int choice;

do
{
    printf("\n===== MENU =====\n");
    printf("1. Prove matrix addition is commutative\n");
    printf("2. Prove matrix multiplication is not commutative\n");
    printf("3. Exit\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            matrixAddition();
            break;

        case 2:
            matrixMultiplication();
            break;

        case 3:
            printf("Program terminated.\n");
            break;

        default:
            printf("Invalid choice! Please enter 1, 2 or 3.\n");
    }

} while (choice != 3);

return 0;
}

```

Test Cases

Test Choice		Input	Expected Result
1	1	A = 1 2; 3 4, B = 5 6; 7 8	A+B = B+A = [6 8; 10 12]
2	2	A = 1 2; 3 4, B = 5 6; 7 8	AB != BA
3	2	Invalid dimensions for both products	Displays dimension error
4	3	—	Program terminates
5	5	—	Invalid choice message

For **Test Case 2:**

```

AB =
19  22
43  50

```

```
BA =  
23    34  
31    46
```

Therefore,

```
AB != BA  
Hence, matrix multiplication is NOT COMMUTATIVE.
```

Detailed Evaluation of Main Test Cases

TC-1 — Matrix Addition



```
(kali@kali)-[~/Desktop]  
$ ./a.out  
  
===== MENU =====  
1. Prove matrix addition is commutative  
2. Prove matrix multiplication is not commutative  
3. Exit  
Enter your choice: 1  
Enter number of rows and columns: 2 2  
Enter elements of Matrix A:  
1 2  
3 4  
Enter elements of Matrix B:  
5 6  
7 8  
  
A + B:  
6      8  
10     12  
  
B + A:  
6      8  
10     12  
  
A + B = B + A  
Hence, matrix addition is COMMUTATIVE.
```

TC-2 — Matrix Multiplication

```
===== MENU =====
1. Prove matrix addition is commutative
2. Prove matrix multiplication is not commutative
3. Exit
Enter your choice: 2
Enter rows and columns of Matrix A: 2 2
Enter rows and columns of Matrix B: 2 2
Enter elements of Matrix A:
1 2
3 4
Enter elements of Matrix B:
5 6
7 8

AB:
19    22
43    50

BA:
23    34
31    46

AB  $\neq$  BA
Hence, matrix multiplication is NOT COMMUTATIVE.
```